

EXERCISE SHEET 13

GROUP AND RING THEORY

MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

13.1. Compute the invariants and write down the structures of the abelian groups with generators $x_1, \dots, x_n$ subject to the following relations:

- (a) $n = 2, x_1 + x_2 = 0$;
- (b) $n = 3, 3x_1 - 2x_2 = 0, x_1 + x_3 = 0$, and $-x_1 + 3x_2 + 2x_3 = 0$;
- (c) $n = 3, 2x_2 - x_3 = 0, -3x_1 + 8x_2 + 3x_3 = 0$, and $2x_1 - 4x_2 - x_3 = 0$.

EXTRA QUESTIONS

13.2. Let R be a principal ideal domain, and let M be an R -module. Show that

$$\text{Tor } M := \{x \in M \mid x \text{ is torsion}\}$$

is a submodule of M .